

ENRIQUE GARCIA RIVERA

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EDUCATION

Washington University in St. Louis, McKelvey School of Engineering

Expected: May 2027

B.S. Mechanical Engineering, Minors in Robotics and Mechatronics

GPA: 3.64 / 4.00

Relevant Coursework: Control Systems (MATLAB/Simulink), Numerical Methods and Matrix Algebra, Fluid Mechanics, Heat Transfer, Solid Mechanics, Vibrations, Electrical and Electronic Circuits for Mechanical Engineers

SKILLS

- **Engineering Competencies:** System monitoring and troubleshooting, sensor instrumentation, telemetry style data analysis, control systems fundamentals, experimental testing, technical documentation.
- **Software and Tools:** MATLAB, Python, C++, Excel, SolidWorks, ANSYS Mechanical, embedded systems (Teensy and Arduino), oscilloscope and multimeter testing.

EXPERIENCE

WashU VTOL – Avionics & Systems Integration Lead, St. Louis, MO

Sep 2025 – Present

- Designed and integrated avionics architecture for a prototype VTOL aircraft including flight controller, onboard computer, GPS, IMU, telemetry radios, and power distribution systems.
- Configured navigation sensors and telemetry links to support reliable communication between onboard systems and ground monitoring tools.
- Conducted staged subsystem validation through bench testing and system monitoring to verify sensor functionality and data communication.
- Collected and analyzed navigation sensor data to evaluate system stability and confirm correct operation of integrated avionics components.
- Diagnosed integration issues between sensors, computing hardware, and communication systems during prototype development.
- Documented avionics architecture, wiring layouts, and test procedures for repeatable integration and troubleshooting.

Robotics Lab, UMKC – Research Intern (Grant Funded), Kansas City, MO

May 2025 – Aug 2025

- Built an embedded sensing platform synchronizing IMU and EMG sensor streams at 1 kHz for robotics experiments.
- Evaluated measurement reliability by characterizing sensor noise, bias drift, and timing jitter across experimental trials.
- Implemented an Extended Kalman Filter in MATLAB to fuse multiple sensor measurements and estimate motion state variables.
- Developed Python scripts to process and visualize experimental data collected during testing.
- Conducted experimental trials to evaluate system behavior and validate sensor fusion performance.
- Presented project results as a research poster at the IEEE Body Sensor Networks Conference (BSN) 2025.

WashU Design Build Fly – Aerodynamics Engineer, St. Louis, MO

Sep 2024 – Jan 2026

- Evaluated aircraft wing geometries using aerodynamic analysis tools to study lift and drag performance across design configurations.
- Performed structural finite element analysis in ANSYS to verify wing spar load capacity and structural safety margins.
- Designed mechanical components including payload release mechanisms using CAD modeling and additive manufacturing methods.
- Documented design procedures and testing configurations to support repeatable aircraft builds and knowledge transfer across the team.

Washington University McKelvey School of Engineering – Course Assistant, St. Louis, MO

Aug 2025 – Dec 2025

- Supported MEMS 2000 Electrical and Electronic Circuits for Mechanical Engineers through laboratory sessions and instrumentation troubleshooting.
- Assisted students with oscilloscope measurements, circuit debugging, and experimental setup.

FedEx Corporation – Material Handler, Kansas City, MO

Jun 2024 – Aug 2024

- Maintained high throughput while following strict safety procedures in a fast paced logistics environment.
- Demonstrated reliability, attention to detail, and teamwork while operating under time sensitive operational conditions.

TECHNICAL PROJECTS

- **Hyperion Orbital Physics Simulator (C++):** Developed a six degree of freedom orbital dynamics simulator implementing the J2 perturbation model to study spacecraft motion and orbital behavior.
- **Flight Navigation EKF System (Embedded C++):** Implemented a twelve state Extended Kalman Filter combining GPS, barometer, and IMU measurements to estimate navigation states for autonomous systems.
- **Sentinel Prototype (C++ and Python):** Designed and built a robotic sensing system integrating embedded control and sensor feedback for experimental tracking applications.